Exercise Sheet #6: Transformer and Transients

Discussion: Nov 24, 2025

Exercise 1: Ideal Transformer

A 480-V to 2400-V step-up ideal transformer delivers 50 kW to a resistive load. Calculate:

a) the turns ratio

b) the primary current

c) the secondary current

Solution

a) Turns ratio (primary to secondary):

$$\alpha = \frac{V_P}{V_S} = \frac{480\,\text{V}}{2400\,\text{V}} = 0.2$$

b) Primary current:

$$I_P = \frac{P}{V_P} = \frac{50000\,\text{W}}{480\,\text{V}} \approx 104.17\,\text{A}$$

c) Secondary current:

$$I_S = \frac{P}{V_S} = \frac{50000\,\text{W}}{2400\,\text{V}} \approx 20.83\,\text{A}$$

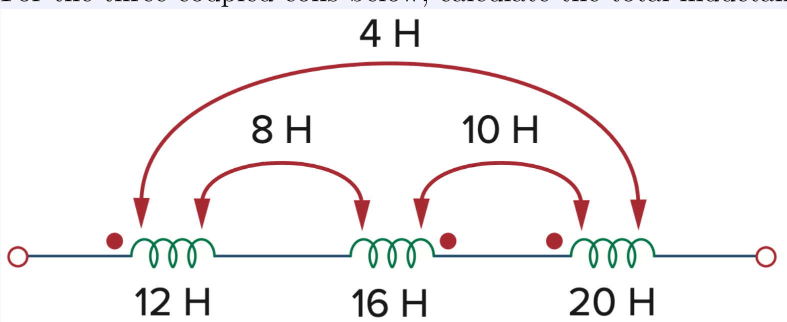
Check:

$$\frac{I_S}{I_P} \approx 0.2$$

Remark: For AC, one would simply consider RMS values.

Exercise 2: Mutual Inductance

For the three coupled coils below, calculate the total inductance.



Background: The polarity of the mutual voltage is defined by the orientation or particular way in which the two coils are physically wound. It can be determined by applying Lenz's law in conjunction with the right-hand rule. In circuit diagrams, the polarity is often shown by the dot convention: A dot is placed in the circuit at one end of each of the two magnetically coupled coils to indicate the direction of the magnetic flux if current enters that dotted terminal of the coil. The convention then reads: If a current enters the dotted terminal of one coil, the reference polarity of the mutual voltage in the second coil is positive at the dotted terminal of the second coil.

Solution

$$\text{Given: } \begin{cases} L_1 = 12\ \text{H}, \\ L_2 = 16\ \text{H}, \\ L_3 = 20\ \text{H}, \\ M_{12} = 8\ \text{H}, \\ M_{23} = 10\ \text{H}, \\ M_{13} = 4\ \text{H}. \end{cases}$$

The total inductance of a system of magnetically coupled coils connected in series depends not only on the self-inductances of the coils, but also on the mutual inductances between them.
When two coils are connected such that the magnetic flux produced by one *aids* the flux in the other, the mutual inductance contributes *positively.* If the fluxes *oppose* each other, the mutual inductance contributes *negatively.*
In the given circuit:

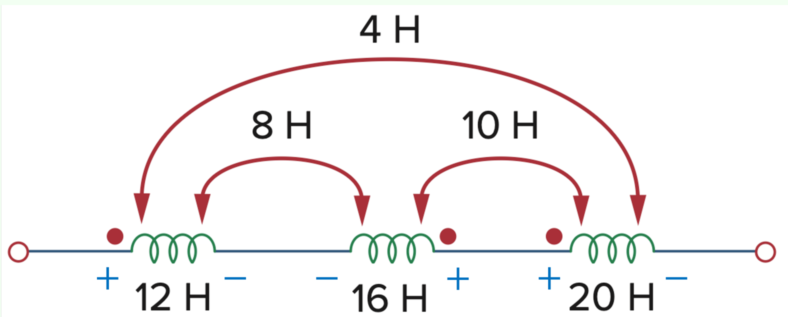
1. The dots indicate the polarity of the induced voltage in each coil.
2. The connection pattern shows that:
   (a) M_{12} and M_{23} are *opposing* (the current enters the dotted terminal of one coil and leaves the dotted terminal of the other).
   (b) M_{13} is *aiding* (the current enters both dotted terminals).

For three series-connected coupled coils, the equivalent inductance is given by:

$$L_{\text{eq}} = L_1 + L_2 + L_3 + 2(\pm M_{12} \pm M_{23} \pm M_{13}),$$

where the signs depend on whether the mutual coupling is aiding or opposing.
From the polarity information in the diagram, we have:

$$L_{\text{eq}} = L_1 + L_2 + L_3 + 2(M_{13} - M_{12} - M_{23}) = (12 + 16 + 20 + 2(4 - 8 - 10))\ \text{H} = 20\ \text{H}$$



Exercise 3: RC-Discharge

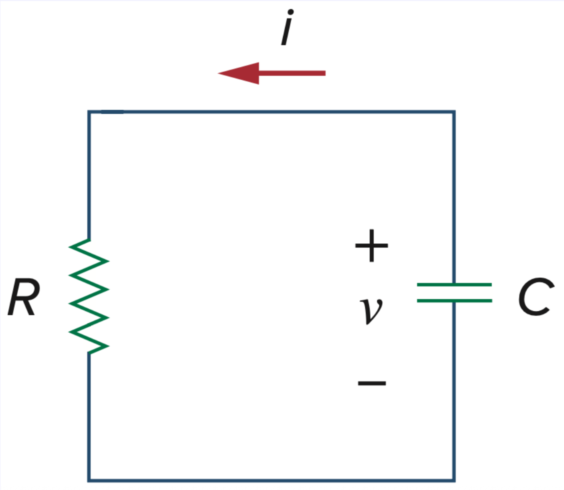
In the circuit shown below, for $t > 0$

$$v(t) = 56e^{-200t}\,\mathrm{V}, \qquad i(t) = 8e^{-200t}\,\mathrm{mA}.$$

a) Find the values of R and C.

b) Calculate the time constant τ.

c) Determine the time required for the voltage to decay to half its initial value at $t = 0$.



Solution

a) For a single series R–C discharge, the capacitor voltage has the form

$$v(t) = V_0 e^{-t/(RC)}.$$

Comparing $v(t) = 56e^{-200t}$ with $V_0 e^{-t/(RC)}$ gives

$$\frac{1}{RC} = 200 \quad \Longrightarrow \quad RC = 5\,\mathrm{ms}.$$

The capacitor current is related to the capacitor voltage by

$$i(t) = C\frac{dv(t)}{dt}.$$

Differentiate $v(t)$:

$$\frac{dv}{dt} = -200 \cdot 56e^{-200t}\,\mathrm{V/s} = -11200e^{-200t}\,\mathrm{V/s}.$$

Hence

$$C = \frac{i(t)}{dv/dt} = \frac{0.008e^{-200t}}{-11200e^{-200t}}.$$

Taking the positive magnitude,

$$C = \frac{0.008}{11200}\,\mathrm{F} \approx 714.3\,\mathrm{nF}.$$

$$R = \frac{RC}{C} = \frac{0.005}{714.3 \times 10^{-9}}\,\Omega = 7\,\mathrm{k}\Omega$$

(Alternatively, $R = v(t)/i(t) = 56/0.008\,\Omega = 7\,\mathrm{k}\Omega$.)

b) The time constant is

$$\tau = RC = 5\,\text{ms}.$$

c) We require $v(t_{1/2}) = \frac{1}{2}V_0$. For $v(t) = V_0 e^{-t/\tau}$,

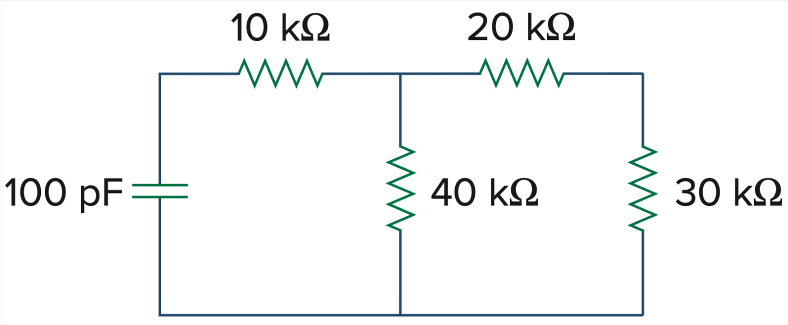
$$\frac{1}{2} = e^{-t_{1/2}/\tau} \quad \Longrightarrow \quad t_{1/2} = \tau \ln 2.$$

Thus

$$t_{1/2} = (0.005 \cdot \ln 2)\,\text{s} \approx 3.466\,\text{ms}.$$

Exercise 4: Time Constant

Determine the time constant for the circuit below.



Solution

Find the Thevenin resistance:

$$R_{\text{th}} = 10\text{ k}\Omega + (40\text{ k}\Omega \parallel (20\text{ k}\Omega + 30\text{ k}\Omega)) = 10\text{ k}\Omega + \frac{40 \cdot 50}{40 + 50}\text{ k}\Omega = \frac{290}{9}\text{ k}\Omega$$

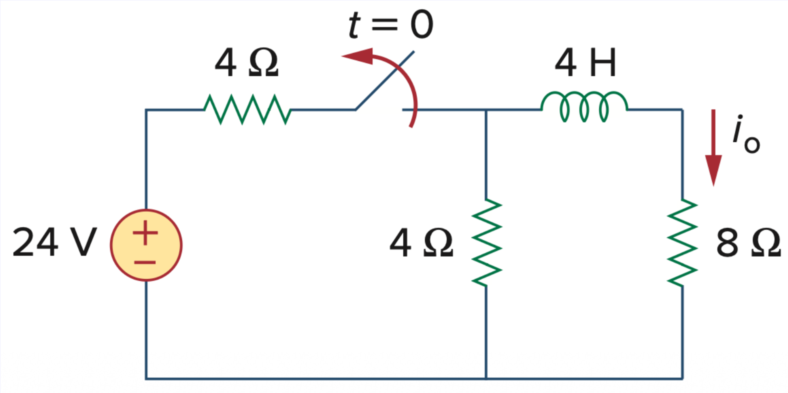
Compute the time constant:

$$\tau = R_{\text{th}}C = \left(\frac{290}{9} \times 10^3\ \Omega\right)\left(100 \times 10^{-12}\text{ F}\right) = \frac{29000}{9} \times 10^{-9}\text{ s}$$

$$\tau \approx 3.22\ \mu\text{s}$$

Exercise 5: Inductor Current Decay

For the circuit below, find i_o for $t > 0$.



Solution

The switch is closed for $t < 0$. Therefore, the inductor is in DC steady state and behaves as a short circuit.
The node after the left 4Ω resistor sees the parallel of 4Ω and 8Ω:

$$R_{\text{eq}} = \frac{4 \cdot 8}{4 + 8} = \frac{8}{3}\ \Omega.$$

The node voltage is

$$v(0^-) = 24\,\frac{R_{\text{eq}}}{4 + R_{\text{eq}}} = 24\,\frac{8/3}{4 + 8/3} = 9.6\ \text{V}.$$

Thus the inductor current at $t = 0^-$ is

$$i_L(0^-) = \frac{v(0^-)}{8} = \frac{9.6}{8} = 1.2\ \text{A}.$$

Therefore,

$$i_L(0^+) = 1.2\ \text{A}.$$

After the switch opens at $t = 0$, the source is disconnected. The top node is connected only to a 4Ω resistor to ground and the inductor with $L = 4\,\text{H}$ in series with an 8Ω resistor.
Applying KCL at the top node (currents downward positive):

$$\frac{v}{4} + i_L = 0$$

Hence,

$$v = -4i_L.$$

The right branch voltage is also

$$v = 8i_L + L\frac{di_L}{dt} = 8i_L + 4\frac{di_L}{dt}.$$

Equate the two expressions for v:

$$-4i_L = 8i_L + 4\frac{di_L}{dt}$$

Divide by 4:

$$\frac{di_L}{dt} + 3\,i_L = 0.$$

Thus,

$$i_L(t) = i_L(0^+)\,e^{-3t} = 1.2\,e^{-3t}.$$

$$\boxed{i_o(t) = 1.2\,e^{-3t}\ \text{A}, \quad t > 0.}$$

Exercise 6: Energy Dissipation during Transient

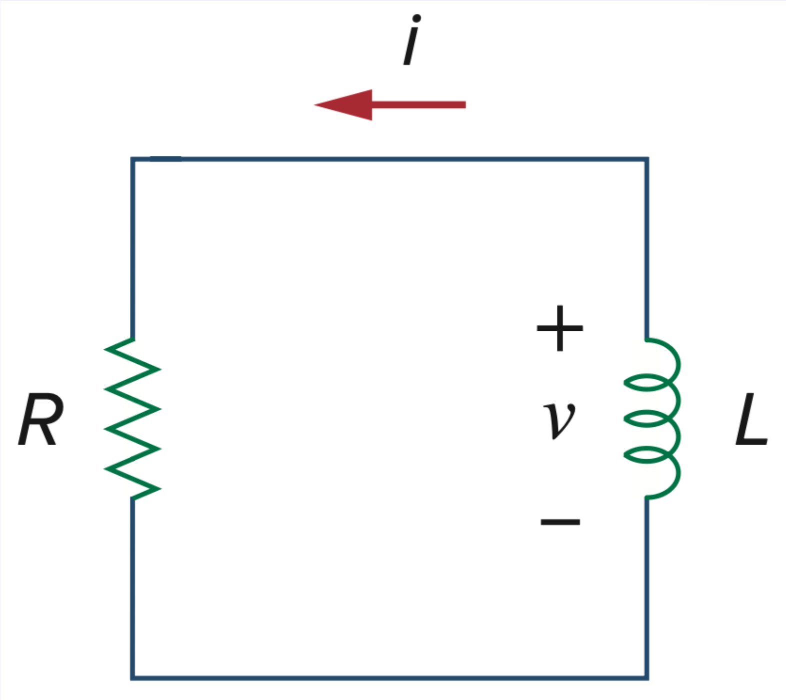
In the circuit below,

$$v(t) = 80e^{-10^3 t}\ \text{V}, \qquad t > 0,$$

$$i(t) = 5e^{-10^3 t}\ \text{mA}, \qquad t > 0.$$

a) Find $R,\ L,\ \tau$.

b) Calculate the energy dissipated in the resistance for $0 < t < 0.5\,\text{ms}$.



Solution

a) The current has the form $i(t) = I_0 e^{-t/\tau}$. Comparing exponents,

$$\frac{1}{\tau} = 1000 \quad \Longrightarrow \quad \tau = 1\,\text{ms}.$$

Since $v(t)$ is also the voltage across the resistor,

$$R = \frac{v(t)}{i(t)} = \frac{80e^{-1000t}}{5 \times 10^{-3} e^{-1000t}} = 16\,\text{k}\Omega.$$

Then

$$L = \tau R = (1 \times 10^{-3})(1.6 \times 10^{4}) = 16\,\text{H}.$$

b) The energy dissipated in R from $t = 0$ to $t = t_1$ is

$$W = \int_0^{t_1} i^2(t)\, R\, dt, \qquad t_1 = 0.5\,\text{ms}.$$

Thus,

$$W = R \int_0^{t_1} (5 \times 10^{-3})^2 e^{-2000t}\, dt = R\,(25 \times 10^{-6}) \int_0^{t_1} e^{-2000t}\, dt.$$

Evaluate the integral:

$$\int_0^{t_1} e^{-2000t}\, dt = \left[-\frac{1}{2000} e^{-2000t} \right]_0^{t_1} = \frac{1}{2000}\left(1 - e^{-2000t_1}\right).$$

Therefore,

$$W = R\,(25 \times 10^{-6})\frac{1}{2000}(1 - e^{-2000t_1}).$$

Substitute $R = 1.6 \times 10^4$ and $2000t_1 = 2000(5 \times 10^{-4}) = 1$:

$$W = (1.6 \times 10^4)(25 \times 10^{-6})\frac{1}{2000}(1 - e^{-1}) = 1.26424 \times 10^{-4}\,\mathrm{J} \approx 0.126\ \mathrm{mJ}.$$